

EDUCATION

University of Florida

Master of Science, Electrical and Computer Engineering

Gainesville, FL

Aug 2022 - May 2024

- Relevant Courses : Fundamentals of Machine Learning, Parallel Computer Architecture, Neural Networks and Deep Learning, Agile Project Management, Computer Communications, IoT Security and Privacy.

EXPERIENCE

Walmart | Stack: ML, Applied AI, Generative AI, NLP, Search Relevance, Information Retrieval, Ranking Systems

Bentonville, AR

Data Scientist - III

Oct 2024 -Present

Project: PT Prediction Improvement using Google SEO Data

- Improved Product Type (PT) prediction recall by integrating Google SEO query data into existing QUMT training pipelines, achieving a **7.93% increase in recall** for newly emerging and low-coverage queries.
- Developed scalable PySpark pipelines for SEO query processing, engagement analysis, and training data generation across large-scale search datasets.
- Enhanced PT coverage using sibling expansion and semantic query understanding techniques to improve search relevance and retrieval quality.
- Evaluated experimental models against production baselines using offline ranking metrics such as recall and nDCG.

Project: Query-PT Classifier Enhancement using New Labeling Methodology

- Developed a new labeling methodology for Query-PT classification using engagement signals, SEO data, and LLM-generated predictions to improve semantic query understanding.
- Achieved a **14-point increase in recall** and **0.4 improvement in precision** through optimized query-product type mapping and classifier enhancements.
- Performed large-scale model evaluation, threshold optimization, and production baseline comparisons to validate search relevance improvements.
- Collaborated with cross-functional search relevance teams to support experimentation, ranking signal development, and deployment of ML-driven search improvements.

University of Florida | Stack: Genomic Data Analysis, Python, CUDA, XGBoost, GPN-MSA

Gainesville, FL

Graduate Research and Student Mentor

Aug 2024 - Oct 2024

- Applied advanced machine learning techniques, including supervised learning and deep learning, to analyze complex non-coding DNA sequences, achieving 88.2% accuracy in genomic annotation
- Developed machine learning models with a focus on scalable architecture and robust design patterns.
- Utilized GPN-MSA transformer models and XGBoost, paralleling LLMs' methodologies, to refine predictive models and increase classification precision by 20%, demonstrating cutting-edge applications in ensemble learning and shrinkage methods

Accenture | Stack: Python, MySQL, CI/CD, SDLC

Hyderabad, India

Associate Software Engineer

July 2021 - July 2022

- Managed incidents for BMW project, achieving 95% compliance with SLA targets; led root cause analysis, identified key issues and implemented permanent solutions that decreased recurring issues by 50%.
- Utilized Git for version control and participated extensively in the full software development lifecycle.
- Developed and maintained software solutions using Java and Python, consistently applying best practices in programming and system design.

Engineers Hub | Stack: Android Studio, Node.js, React, UI/UX,

Hyderabad, India

Mobile App Development Intern

June 2019 -July 2019

- Designed the concept and initiated the frontend design phase of the Hand Out mobile application, aimed at reducing food waste and supporting orphanages.
- Collaborated closely with the team to establish project milestones and deliverables.
- Analyzed industry-standard tools like Android Studio and version control systems like Git to facilitate efficient development processes.

SKILLS

Programming Languages: C/C++, Python, R, SQL, Java**Data Analysis & Big Data:** Pandas, NumPy, SciPy, Statistics, Excel, Apache Spark, PySpark, Apache Hadoop**Data Visualization:** Power BI, Tableau, Matplotlib, Seaborn**Machine Learning & AI:** Scikit-Learn, XGBoost, TensorFlow, PyTorch, Keras, OpenCV, Supervised Learning, Unsupervised Learning, ANN, CNNs, RNNs, LSTM, GANs, Autoencoders, Transfer Learning, LangChain, LLMs, Generative AI**Cloud & Distributed Computing:** AWS (EC2, S3, SageMaker, Amplify, Lambda, Elastic Beanstalk), Microsoft Azure, GCP**Systems & Tools:** Linux/Unix, OpenMP, CUDA, OpenCL, GPUs, CPU Architecture, CI/CD, JIRA, Anaconda, JupyterHub, MATLAB

PROJECTS

Sentiment Analysis Using LSTM / [GitHub](#) / Python, TensorFlow/Keras, AWS SageMaker, NLP

- Conducted comprehensive sentiment analysis on movie reviews, employing Natural Language Processing (NLP) with an LSTM-based model to extract and understand user sentiments effectively.
- Enhanced analysis quality by removing noise with NLTK and employed visualization tools to identify key sentiment indicators, which helped pinpoint areas for model improvement.
- Successfully developed and optimized the LSTM model on AWS SageMaker, achieving an impressive 96.32% accuracy and an F1 Score of 94%, demonstrating the model's effectiveness in classifying positive and negative sentiments efficiently.

Brand Logo Classification / [GitHub](#) / Python, Tensorflow, Jupyter, VGG16

- Implemented a Convolutional Neural Network (CNN) classification system with transfer learning capabilities to categorize brand logos, accuracy was enhanced by 20% and shortened classification time by 60%

- Achieved an accuracy of 94.62% with VGG16 through hyperparameter tuning, including early stopping and the implementation of Adam optimizer.

THE THIRD EYE / [GitHub](#) / *Python, C++, VNC Viewer, Raspberry pi, OpenCV, Resnet50*

- Developed "The Third Eye," a Raspberry Pi-based smart cane for visually impaired individuals.
- Integrated voice-based assistance, object detection, and GPS technology to enable independent travel.
- Empowered users with real-time feedback, navigation guidance, and enhanced situational awareness.

Movie Recommendation System / [GitHub](#) / *Python, TensorFlow/PyTorch, Streamlit, Data Analytics*

- Engineered a dynamic Movie Recommendation System using Python and TensorFlow/PyTorch, achieving a 95% accuracy in predicting user preferences through collaborative and content-based filtering techniques.
- Developed and deployed an interactive web application using Streamlit, enabling real-time, user-friendly interactions that allowed users to receive personalized movie suggestions.

Publications/Patents

- **THE THIRD EYE:** Invented THE THIRD EYE, an AI-powered assistive tool for the visually impaired, integrating Artificial Intelligence, Computer Vision and deep learning methodologies (Indian Patent Office, 202141010185).